

CSE 112 – Lab #8 Corporate Sales – Vectors, Classes, and Storing Objects

Reference Programs 17-7 thru 17-10 and the Product.h code on page 1084

Implement the Product class on page 1084 in a header file Products.h adding the attributes shown below. Create instances of the class, populate the objects with the data provided below from a file called 'productInfo.txt', and store the objects in a vector.

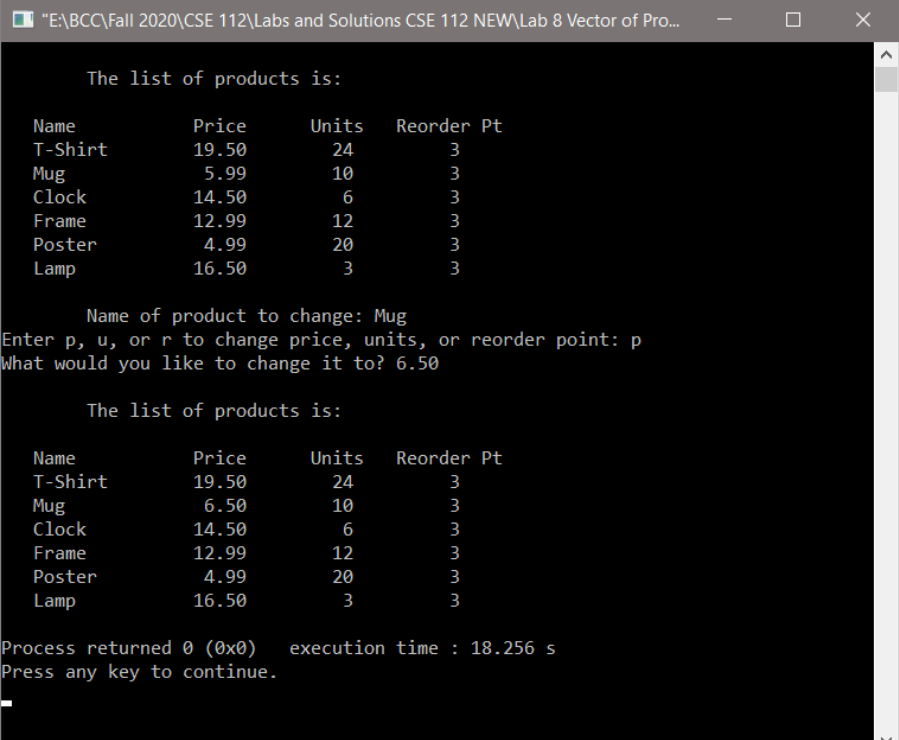
1. Display all of the products and their information (see below) using a function.
2. Prompt the user for the name of the product they would like to change.
3. Prompt the user for the change – 'p' = price, 'u' = units, 'r' = reorder point
4. Update the product information
5. Display all of the product information calling the display function.

```
void display_products(vector<Product> p);
```

If there is a change and it does not persist when the products are displayed and you are using 'auto' in the loop, change to accessing the vector like an array with brackets.

```
for(int i = 0; i < 6; i++)
```

The program will use the class definition provided below in a separate *.h file, and use the member functions as appropriate.

Output Format – tabs and spacing required:


```

E:\BCC\Fall 2020\CSE 112\Labs and Solutions CSE 112 NEW\Lab 8 Vector of Pro...
The list of products is:
Name      Price      Units      Reorder Pt
T-Shirt   19.50       24         3
Mug        5.99        10         3
Clock     14.50        6          3
Frame     12.99       12          3
Poster    4.99        20          3
Lamp      16.50        3           3

Name of product to change: Mug
Enter p, u, or r to change price, units, or reorder point: p
What would you like to change it to? 6.50

The list of products is:
Name      Price      Units      Reorder Pt
T-Shirt   19.50       24         3
Mug        6.50        10         3
Clock     14.50        6          3
Frame     12.99       12          3
Poster    4.99        20          3
Lamp      16.50        3           3

Process returned 0 (0x0)   execution time : 18.256 s
Press any key to continue.

```

// Product class definition for Product.h file

```
class Product
{
    private:
        string name;
        int units;
        double price;
        int reOrderPoint;

    public: // constructor
        Product(string n, double p, int u)
        {
            name = n;
            price = p;
            units = u;
            reOrderPoint = 3;
        }

        void setName(string n)
        { name = n; }

        void setPrice(double p)
        { price = p; }

        void setUnits(int u)
        { units = u; }

        void setReorderPoint(int r)
        { reOrderPoint = r; }

        string getName() const
        { return name; }

        double getPrice() const
        { return price; }

        int getUnits() const
        { return units; }

        int getReorderPoint() const
        { return reOrderPoint; }
};
```

Date for "productInfo.txt"

T-Shirt

19.50

24

Mug

5.99

10

Clock

14.50

6

Frame

12.99

12

Poster

4.99

20

Lamp

16.50

3